Financial Aspect & Target Audience

Hyderabad Metro Route Optimizer

Team Members

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Fare Structure Analysis • Revenue Model • Cost Breakdown

Ridership Estimates • Target Audience Segments

Metro Rider Personas • Use Cases • Value Proposition

1 Fare Structure of Hyderabad Metro

Hyderabad Metro operates on a distance-based slab fare system set by HMRL (Hyderabad Metro Rail Limited). The fare depends ONLY on the total distance travelled on the metro network — not on the number of lines used or the route taken. This means a metro-only path from A to B costs the same regardless of which route the optimizer recommends, unless first/last-mile costs differ.

Official Fare Slabs (Rs. 10 – Rs. 60)

Distance (km)	Fare (Rs.)	Per km (Rs.)
0–2	**Rs. 10**	5.0
2–4	**Rs. 15**	3.8
4–6	**Rs. 20**	3.3
6–8	**Rs. 25**	3.1
8–10	**Rs. 30**	3.0
10–14	**Rs. 35**	2.5
14–18	**Rs. 40**	2.2
18–22	**Rs. 45**	2.0
22–26	**Rs. 50**	1.9
26–30	**Rs. 55**	1.8
>30	**Rs. 60**	—

The minimum fare is Rs. 10 for up to 2 km, and the maximum is Rs. 60 for distances above 30 km. There are 11 fare slabs in total. The per-kilometre cost decreases as distance increases, making longer journeys more economical per kilometre.

2 Revenue Model of the Application

The Hyderabad Metro Route Optimizer is a non-commercial academic project. It does not generate direct revenue. However, understanding the financial dynamics of the metro system it models is essential for evaluating its real-world applicability.

HMRL Revenue Sources

- Ticket Sales

Primary revenue source. Fares range from Rs. 10 to Rs. 60 per ride based on distance slabs.

- Smart Cards & QR Tickets

Pre-loaded smart cards and mobile QR tickets encourage repeat usage and advance payment.

- Station Retail & Advertising

Commercial space rental at stations and advertising on trains/platforms.

- Park-and-Ride Fees

Parking charges at metro stations for commuters who drive to the station.

- Last-Mile Connectivity

Feeder bus services and auto-stands at stations generate auxiliary revenue.

Application Cost Model (This Project)

- Development

Open-source tools (React, TypeScript, Vite, Tailwind CSS) — zero licensing cost.

- Hosting

Supabase free tier for database/auth; static hosting for the frontend — zero cost.

- Data

Station names and line data from public HMRL sources. Distances and times are simulated — zero data purchase cost.

- Maintenance

Community or academic maintenance. No paid API subscriptions required.

3 Ridership Estimates & Financial Dynamics

The Hyderabad Metro network spans 3 operational lines covering 67.21 km with 56 stations. Understanding ridership patterns helps evaluate the financial viability of the system and the value of a route optimization tool.

Network Length **67.21 km**	Total Stations **56**	Operating Lines **3**
Fare Range **Rs. 10–60**	Operating Hours **06:30–22:00**	Headway **5–8 min**

Financial Impact of Route Optimization

- Fare Savings for Riders

The "cheapest" objective finds routes in the lowest fare slab. For cross-city journeys, choosing a shorter route can save Rs. 5–15 per trip.

- Time Savings = Economic Value

The "fastest" objective minimises travel time. For a daily commuter saving 10 minutes each way, that is 100 hours saved per year — equivalent to over 4 full days.

- Transfer Reduction

The "fewest interchanges" objective reduces platform changes, saving 5 minutes per avoided transfer and reducing the risk of missed connections.

- Informed Fare Decisions

By showing estimated fares upfront, riders can choose between metro, auto, or bus based on total cost, increasing metro adoption for cost-sensitive segments.

4 Cost-Benefit Analysis for Riders

This analysis compares the financial cost of using the Hyderabad Metro versus alternative transport options for typical journey scenarios, using the fare model built into the application.

Journey Type	Metro Fare	Auto/Rickshaw	City Bus	Savings vs Auto
Short (2–4 km)	Rs. 10–15	Rs. 10–15	Rs. 25–45	
Medium (8–14 km)	Rs. 25–35	Rs. 80–140	Rs. 15–25	Rs. 45–105
Long (18–27 km)	Rs. 40–50	Rs. 180–250	Rs. 25–40	Rs. 130–210
Cross-city (25–37 km)	Rs. 55–60	Rs. 250–350	Rs. 35–50	Rs. 190–290

Key insight: The metro is significantly cheaper than auto-rickshaws for all journey types, with savings of Rs. 25–290 per trip. For short journeys, city buses compete on price but not on speed or comfort. The route optimizer helps riders discover these savings by computing exact fares for any station pair.

Annual Savings Estimate for a Daily Commuter

- Assume 2 trips/day, 22 working days/month = 44 trips/month
- Medium journey (8–14 km): Metro Rs. 30 vs Auto Rs. 110 = Rs. 80 saved per trip
- Monthly savings: Rs. 80 × 44 = Rs. 3,520
- Annual savings: Rs. 3,520 × 12 = Rs. 42,240
- The route optimizer makes these savings discoverable and actionable.

5 Target Audience: Metro Riders

The Hyderabad Metro Route Optimizer is designed for metro riders — the people who use or could use the Hyderabad Metro for their daily travel. This section breaks down the target audience into distinct segments, each with specific needs that the application addresses.

Why Metro Riders Need This Tool

- The Hyderabad Metro has 3 lines with 56 stations and 3 interchange points. Choosing the right route is not always obvious.
- Fares depend on distance, not route — but riders rarely know the exact distance or fare before travelling.
- Interchanges add 5+ minutes per transfer. Minimising transfers can save significant time for some journeys.
- Different riders prioritise different things: speed, cost, comfort (fewer transfers), or minimal walking.
- First-time riders and tourists may not know which stations connect to which lines.

6 Audience Segments & Personas

A Daily Commuters

Office workers, IT professionals, and government employees who travel the same route every day.

Needs: Fastest route option • Timetable-based departure times • Consistent and repeatable results

Value: Save 10–20 minutes daily by choosing the fastest or fewest-transfer route.

B Students

College and university students travelling between home, campus, and internships across the city.

Needs: Cheapest route option • Fare estimation before travel • Simple, clear instructions

Value: Save Rs. 25–80 per trip by confirming the lowest fare route before boarding.

C Tourists & Visitors

Visitors to Hyderabad who are unfamiliar with the metro network and need guidance.

Needs: Visual network map • Station search by name • Step-by-step itinerary with transfers

Value: Navigate the entire metro confidently without local knowledge.

D Occasional Riders

People who use the metro occasionally for shopping, medical visits, or social outings.

Needs: Balanced route (time + cost + transfers) • Door-to-door itinerary • Operating hours information

Value: Make informed travel decisions without memorising the network.

E Elderly & Mobility-Limited

Senior citizens and people with mobility concerns who want comfortable journeys.

Needs: Fewest interchanges option • Least walking option • Clear transfer instructions

Value: Minimise physical effort by reducing platform changes and walking distances.

F Cost-Conscious Riders

Riders on tight budgets who compare metro fares with bus and auto-rickshaw costs.

Needs: Cheapest route with exact fare • Comparison with alternative transport • Shortest distance route

Value: Maximise savings by choosing the most economical route every time.

7 Use Cases & Value Proposition

Real-World Use Cases

- Morning Office Commute

A software engineer in Madhapur needs to reach Begumpet by 9:30 AM. The app finds the fastest Blue Line route with exact departure time and fare.

- Student Budget Travel

A student in Miyapur needs to reach Nagole for an internship. The "cheapest" objective shows the lowest fare route (Rs. 55–60) versus Rs. 200+ by auto.

- Weekend Shopping Trip

A family in Kukatpally wants to visit Sultan Bazaar. The "balanced" objective finds a route with 1 transfer at MGBS, balancing time and comfort.

- Tourist City Tour

A visitor staying near Ameerpet wants to explore Charminar area. The app shows the Red Line to MGBS, then Green Line to Sultan Bazaar, with a visual map.

- Medical Appointment

An elderly passenger from LB Nagar needs to reach Gandhi Hospital. The "fewest interchanges" route minimises platform changes for comfort.

Value Proposition Summary

- Save Money

Exact fare estimation helps riders choose the cheapest option, saving Rs. 25–290 per trip.

- Save Time

Fastest-route optimization saves 10–20 minutes per journey compared to guessing.

- Reduce Complexity

Visual map + step-by-step itinerary makes the 56-station network navigable for everyone.

- Personalised Routes

Six optimization objectives let each rider prioritise what matters most to them.

- Zero Cost to User

The app is free to use — no subscription, no ads, no data collection.

Bottom Line

The Hyderabad Metro Route Optimizer transforms a complex 3-line, 56-station network into an accessible, personalised journey planner. For metro riders, it means saving money, saving time, and travelling with confidence — every single trip.